Emergency Demand Response Optimization Using Seoul Energy Simulation Data

1. Research Motivation

Modern societies rely heavily on stable electricity supply. During emergencies such as natural disasters, large-scale blackouts, or energy shortages, available power may become insufficient to satisfy all demands.

As South Korea remains under an armistice rather than a permanent peace treaty, I became interested in how limited energy resources should be allocated during crisis situations. This project explores how optimization techniques can support public safety and social resilience through data-driven energy allocation strategies.

2. Dataset

This study utilized Seoul Energy Simulation Data containing daily and hourly energy generation records from multiple distributed energy resources.

Data Characteristics
- Daily records: 327,328 observations
- Hourly records: 7,042,023 observations
- Analysis period: December 2021 - November 2025

Facility Categories
- Hospital (Priority: 100)
- Fire Station (Priority: 95)
- Subway (Priority: 80)
- Residential (Priority: 70)
- Commercial (Priority: 40)

3. Optimization Model

A Linear Programming (LP) optimization model was developed to maximize social utility under energy shortages.

Objective:
$\max \Sigma(p_i \times x_i)$

where:
- p_i = facility priority score
- x_i = allocated energy

Constraints:
1. Total allocation cannot exceed available energy supply.
2. Allocation cannot exceed facility demand.
3. Negative allocation is not allowed.

The model was implemented using Python and SciPy.

4. Emergency Scenarios

- Normal: 100% supply
- Scenario A: 80% supply
- Scenario B: 60% supply

- Scenario C: 40% supply

5. Results

Objective Achievement:
- Normal: 100.0%
- Scenario A: 87.6%
- Scenario B: 72.2%
- Scenario C: 50.5%

Key Findings:
- Hospitals maintained 100% satisfaction across all scenarios.
- Fire stations maintained 100% satisfaction across all scenarios.
- Subway systems remained fully protected.
- Commercial facilities absorbed most reductions.
- Optimization outperformed proportional allocation.

Scenario C (40% Supply):
Hospital: 100%
Fire Station: 100%
Subway: 100%
Residential: 37%
Commercial: 0%

6. Technical Skills Demonstrated

- Python
- Pandas
- NumPy
- SciPy
- Data Preprocessing
- Time-Series Analysis
- Linear Programming
- Energy System Simulation
- Data Visualization

7. Future Research Direction

- Smart Grid Optimization
- Virtual Power Plant (VPP)
- Demand Response Systems
- Renewable Energy Integration
- Energy Informatics
- Digitalized Energy Systems

8. Conclusion

This project demonstrates how computer science and optimization techniques can be applied to real-world energy challenges. The framework provides a practical approach to emergency energy allocation and resilient infrastructure planning.